

combining RT and immunotherapy in extracranial metastatic melanoma. We also report the impact of timing of RT relative to the start of immunotherapy and its impact on clinical outcomes.

Materials/Methods: Patients with extracranial metastatic melanoma from 2004 to 2013 were obtained from the National Cancer Database. Patients were stratified into two groups: immunotherapy alone and RT plus immunotherapy. Univariate and multivariate analysis was performed using Cox model to determine predictors of OS. Kaplan-Meier method was used to compare OS. Subset analysis was performed to evaluate the impact of irradiation of soft tissue lesions (lung, liver, etc) versus bone lesions. Additionally, the impact of the timing of RT relative to the start of immunotherapy was also studied: RT administered 30 days prior to start of immunotherapy (neoadjuvant RT); RT administered within 30 days before or 30 days after the initiation of immunotherapy (concurrent RT); RT administered 30 days after the initiation of immunotherapy (adjuvant RT). Significance was defined as a P value ≤ 0.05 .

Results: A total of 1675 patients were identified: 1387 received immunotherapy alone and 288 received RT plus immunotherapy. The median follow-up time was 15.9 (0.5–136.8) months. On univariate analysis, factors associated with superior OS were younger age, lower Charlson comorbidity score, treatment at academic facility, private insurance, residence in urban and metro areas, and lack of RT. These factors remained significant on multivariate analysis except for the use of RT; use of RT was not associated with lower OS on multivariate analysis. The median OS was 15.4 (12.5–18.6) in RT plus immunotherapy vs. 19.4 (17.6–21.2) months in immunotherapy alone ($P=0.02$). On subset analysis, RT to bone lesions plus immunotherapy had significantly worse OS than immunotherapy alone: 9.4 (4.7–14.1) vs. 19.4 (17.6–21.2) months ($P<0.01$); however, RT to soft tissue lesions plus immunotherapy had statistically similar OS as immunotherapy alone: 18.9 (14.6–23.2) vs. 19.4 (17.6–21.2) months ($P=0.90$). With respect to RT timing, the median OS was statistically similar between neoadjuvant vs. concurrent vs. adjuvant RT groups: 18.9 (10.6–27.2) vs. 15.4 (10.8–20.1) vs. 15.8 (10.7–20.9) months, respectively ($P=0.12$). However, in patients receiving RT to soft tissue lesions, neoadjuvant RT had superior OS compared to concurrent RT or adjuvant RT: median OS 26.1 (12.9 – 39.2) vs 16.0 (10.8 – 21.2) vs 15.4 (9.2 – 21.6) months ($P < 0.01$). This effect was not observed in the patients receiving RT to bone lesions.

Conclusions: These results suggest that the clinical effect of combining RT and immunotherapy in metastatic melanoma is variable. When RT is delivered to bone lesions, it appears to have a detrimental effect; however, when RT is administered to soft tissue lesions, especially at least 30 days

prior to the start of immunotherapy, it may improve survival. Prospective trials are needed to validate these observations and explore potential underlying mechanisms.

(P61) Early Treatment and Toxicity Outcomes for Uveal Melanomas Treated Using Custom Loaded Eye Plaques



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Introduction: Iodine-125 loaded Collaborative Ocular Melanoma Study plaques can achieve excellent tumor control for patients diagnosed with uveal melanomas. However, our ocular cancer team hypothesized that the use of novel, partially loaded COMS plaques could ease and improve accuracy of plaque placement during treatment of small, posterior tumors while providing equivalent tumor control. Here, we compare our single institutional experience with custom-designed, partially loaded plaques to conventional, fully loaded plaques in regards to treatment and toxicity outcomes.

Materials/Methods: Records of 25 patients treated with custom plaques were compared to 20 patients treated with fully loaded plaques, who had received treatment prior to our institution's adopting the use of these partial plaques. Tumors were matched with regards to location and dimensions as measured by the ophthalmologist. Retrospective analysis of dosing parameters, tumor control and toxicity outcomes was performed.

Results: There were no cancer related deaths, local recurrences or metastases in either cohort at an average follow up of 24 months for patients treated with custom plaques and 60.7 months for patients treated with fully loaded plaques. No statistically significant difference was found in regards to post-operative development of cataracts ($\chi^2 = 0.76$) or radiation retinopathy ($\chi^2 = 0.22$). Patients treated with custom loaded plaques noted significantly less clinical visual loss ($\chi^2 = 0.006$) and were more likely to have vision preserved at $\geq 20/200$ ($\chi^2 = 0.006$).

Conclusion: The treatment of small, posterior uveal melanomas with partially loaded COMS plaques results in equivalent survival and recurrence outcomes as treatment with fully loaded plaques, while exposing the patient to less radiation. Additionally, treatment with partially loaded plaques reduces the incidence of clinically significant visual loss. These promising early results support the use of partially loaded plaques in well-selected patients.